

Amin Charepoo

charepoo.a@northeastern.edu | www.linkedin.com/in/amin-charepoo/ | github.com/AminCharepoo | amincharepoo.github.io/WebPortfolio/

EDUCATION

Northeastern University

Boston, MA

Candidate for Bachelor of Computer Engineering | GPA: **3.96** | Dean's List

Sept 2024 - May 2029

Relevant Courses: Discrete Structures, Computing Fundamentals, Differential Equations and Linear Algebra, Calculus 2

Related Activities: Robotics Club, Virtual Reality Lab Assistant, VR Research, Northeastern Virtual Reality Club

TECHNICAL SKILLS

Programming: C++, Python, C#, Java, JavaScript, HTML/CSS

Embedded Systems: ESP32, Arduino, Raspberry Pi

Tools: AutoCAD, Git, MATLAB, Ubuntu Linux, Unity3D, Microsoft Office

PROJECTS

[Real-Time Cooperative Task Scheduler](#) | C++ | Raspberry Pi Pico | Git | LaTeX

Feb 2026 - Apr 2026

- Designed priority-based task scheduler in C++ demonstrating OOP principles.
- Programmed interrupt-driven architecture using GPIO and hardware timer ISRs with safe shared-state management.
- Implemented five cooperating classes managing LCD display with VIP preemption and cooperative task resumption.
- Wrote technical documentation in LaTeX covering system design, pseudocode, and implementation analysis.

[Autonomous Line-Following Robot](#) | Arduino | C++ | Sensors | Git | Embedded Systems

Sept 2025 - Dec 2025

- Programmed an autonomous robot using Arduino C++ to process sensor data and control motors for accurate path-following.
- Developed **modular and maintainable** code with real-time logic for reliable hardware-software integration.
- Calibrated and tested multiple sensors to ensure **responsive and repeatable** robot performance.
- Used **Git** for version control and documented system design to support collaboration and future development.
- Won **first place** in end-of-year competition against three teams.

[Spotify Controller Hub](#) (Personal Project) | C++ | ESP32 | Git | Spotify API

Sept 2025 - Nov 2025

- Designed and built an **embedded** controller that manages hardware inputs to control Spotify playback.
- Programmed an **ESP32** in C++ to handle Wi-Fi communication, API requests, and hardware input processing.
- Created and tested circuit connections for **reliable input handling** and Wi-Fi connectivity.
- Enabled seamless hands-free Spotify control, reducing the need to switch applications and improving **workflow efficiency**.

EXPERIENCE

Northeastern University Virtual Reality Experience

Boston, MA

Research Assistant | C# | Unity3D

Jan 2025 - Aug 2025

- Investigated how haptic and visual feedback interact in Virtual Reality through user testing of a Unity3D simulation programmed in C#.
- Collaborated with a co-researcher to conduct **20+** participants.
- Analyzed findings and shared results through discussion materials and summaries with a team of **8** researchers.

Northeastern University Virtual Reality Immersive Media Lab

Boston, MA

Work Study Associate | C# | Git | Unity3D

May 2025 - June 2025

- Supported VR projects and assisted students and faculty with equipment setup and troubleshooting.
- Collaborated with a team of three to develop a virtual reality story experience in Unity3D using C#.
- Managed version control and collaborative workflows through **Git**.

TechHunt PCs

Boston, MA

Business Owner

Sept 2021 - Jan 2025

- Founded and operated a small business specializing in building, repairing, and selling custom PCs.
- Built and refurbished over **60** computer systems, ensuring reliable hardware operation and performance.
- Diagnosed and resolved hardware and software issues across diverse PC systems, earning **20+** five-star reviews.

ACTIVITIES AND LEADERSHIP

Northeastern Robotics — Intro to Robotics Teacher

(2026-Present)

Northeastern Virtual Reality — Executive Board, Project Lead

(2024-Present)

Boston Ruhi Institute — Volunteer Coordinator, Junior Youth Spiritual Empowerment Program Collaborator

(2018-Present)